THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINSTRATION AND LOCAL GOVERNMENT

KASULU TOWN COUNCIL

FORM TWO PRE-MOCK EXAMINATION

CODE: 041 **BASIC MATHEMATICS**

Time: 2:30 Hours **Tuesday, 20.05.2025**

INSTRUCTIONS

1. This paper consists of **ten (10) compulsory questions**.
2. Show clearly all the working and answers
3. All working must be in **blue** or **black** ink except drawings which must be in pencil.
4. Mathematical tables or graph papers may be used
5. Cellular phones and calculators are not allowed in the examination room.
6. Write your examination number on every page of your answer sheet.

MsomiBora.com

1. (a) Find the product of GCF and LCM of the numbers 90 and 240.

(b) If 500.005 = A x 10^n, where $1 \leq 4 < 10$ and n is an integer. Determine A and n.

2. (a) Simplify the expression $\frac{2^a \, x \, 8^{a+3}}{16^{a-2}}$ completely.

(b) Write the function $\frac{31}{8}$ as mixed number.

3. (a) A cylindrical tank of diameter 140 cm contains water to a height of 2.2m. Calculate the volume of the water in liters.

(b) A person borrows Tsh 600,000/= for a period of 6 years at 20% simple interest per annum. Calculate the total amount the person will pay back after 6 years.

4. (a) If 2x + 70 ° and $3x + 20°$ are supplementary angles, determine the value of X

(b) If a triangle has two equal sides of length x cm each and the length of the remaining side is one quarter of the total length of the two congruent sides, write down an equation that represents the perimeter of this triangle.

5. (a) Determine the value of x that satisfies the equation $\frac{x+10}{x-4} = 3$.

(b) Given that $x^2 - 6x + m$ is a perfect square. Find the values of M.

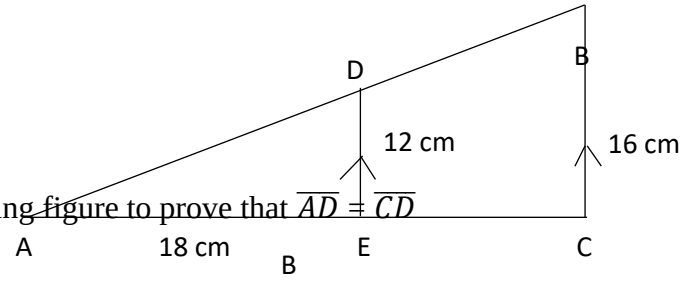
5. (a) Find the equation of a line through the point (2, -2) crossing y-axis at the same point as the line whose equation is $y = \frac{5}{2}x - 5$.

(b) Find the coordinates of the image of the point A (5,2) under a reflection on the line y= 0.

6. (a) Given that $(2^{a-1})(3^{b+1}) = (3^9)(2^8)$. Find $\frac{a-b}{b}$.

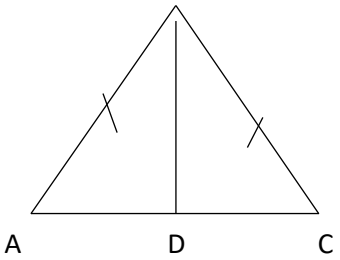
(b) If $\log_{10} 2 = 0.3010, \log_{10} 3 = 0.4771,$ evaluate $\log_{10} 0.6$.

8. (a) In the following figure BC=16 cm, DE= 12 cm, and AE= 18 cm. Calculate the length EC.



(b) Use the following figure to prove that $\overline{AD} = \overline{CD}$

B



9. (a) The lengths of three sides of a right-angled triangle are (x-1) cm, (x-8) cm, and x. Find the value of x.

(b) If tan tan $Q = {}^{5}/_{12}$. Find the value of $\frac{SinQ-2CosQ}{1+SinQ}$.

10. (a) In a class of 45 students, some study physics or chemistry or both. If 23 students study physics, 33 study chemistry and 10 students do not study neither physics nor chemistry, Find the number of students who study both physics and chemistry.

(b) The following table shows marks scored by form two students in a Mathematics test at Mwandiga Secondary School

Marks %	40	45	50	55	60	65	70
No. of students	6	8	13	5	9	4	5

(a) What was the lowest mark

(b) What marks was scored by most students?

(c) If 50% was the pass mark, how many students passed the test?

(d) Find the total number of the students did the test

(e) What was the percentage of the students who scored at least 50%?